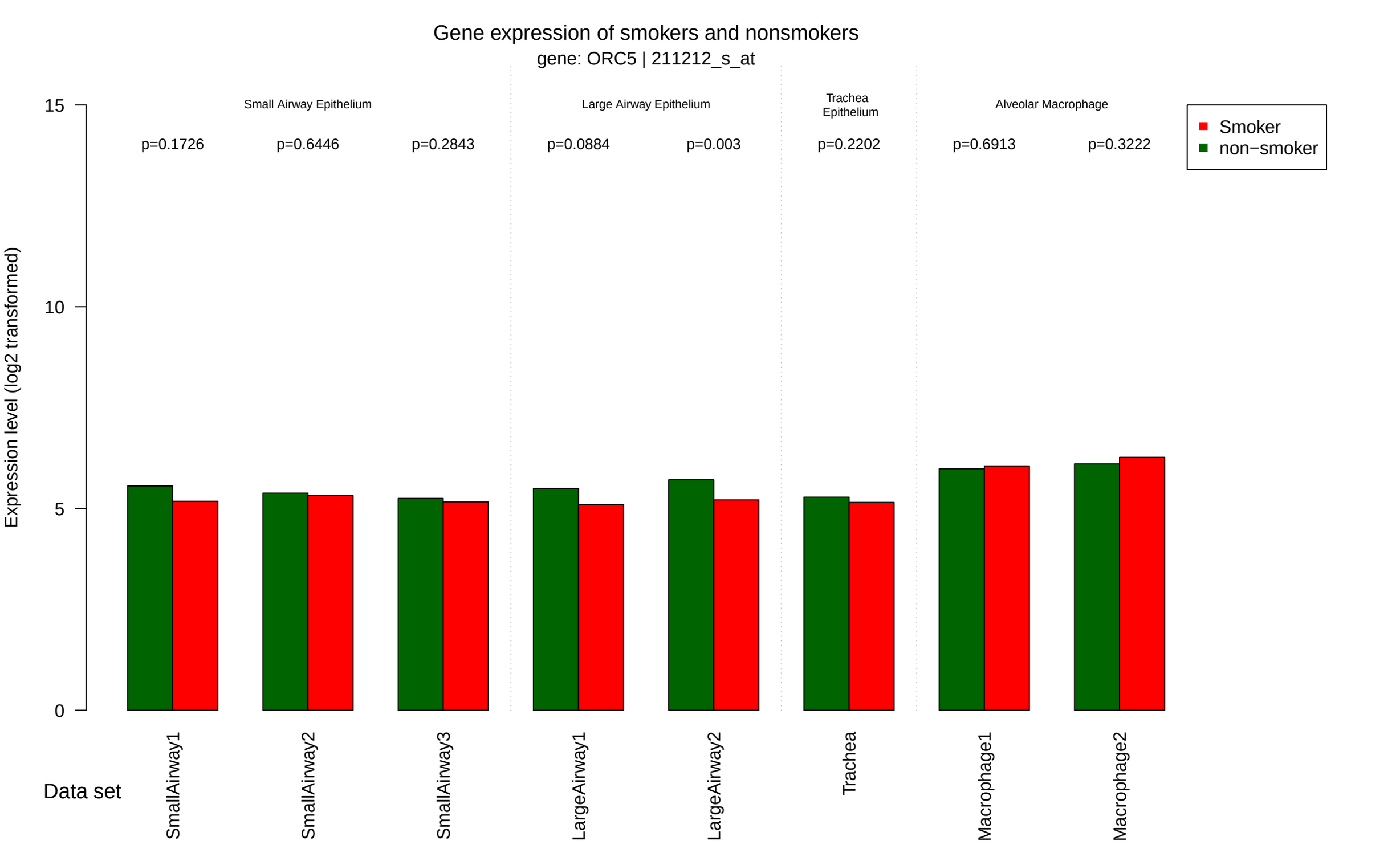
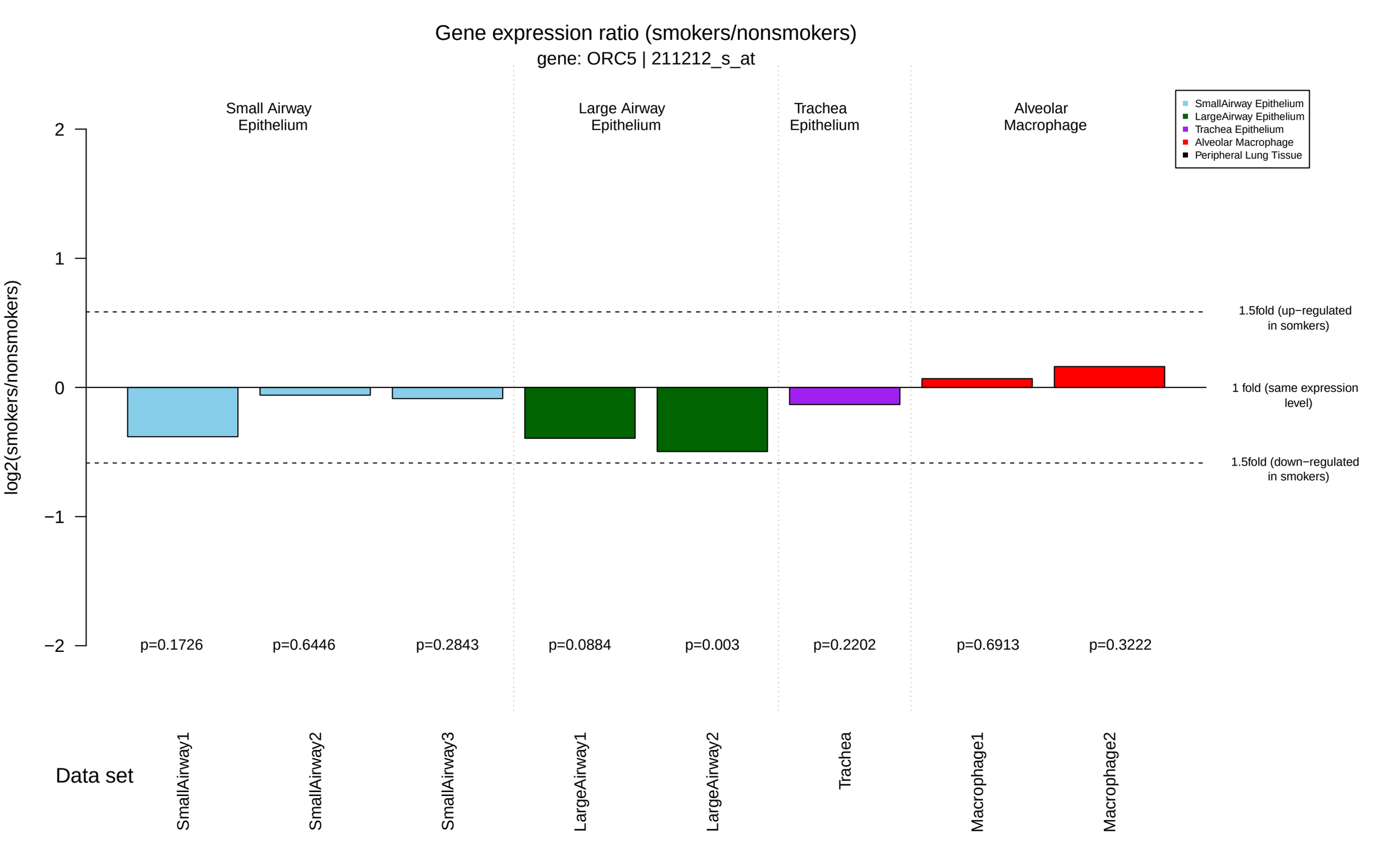
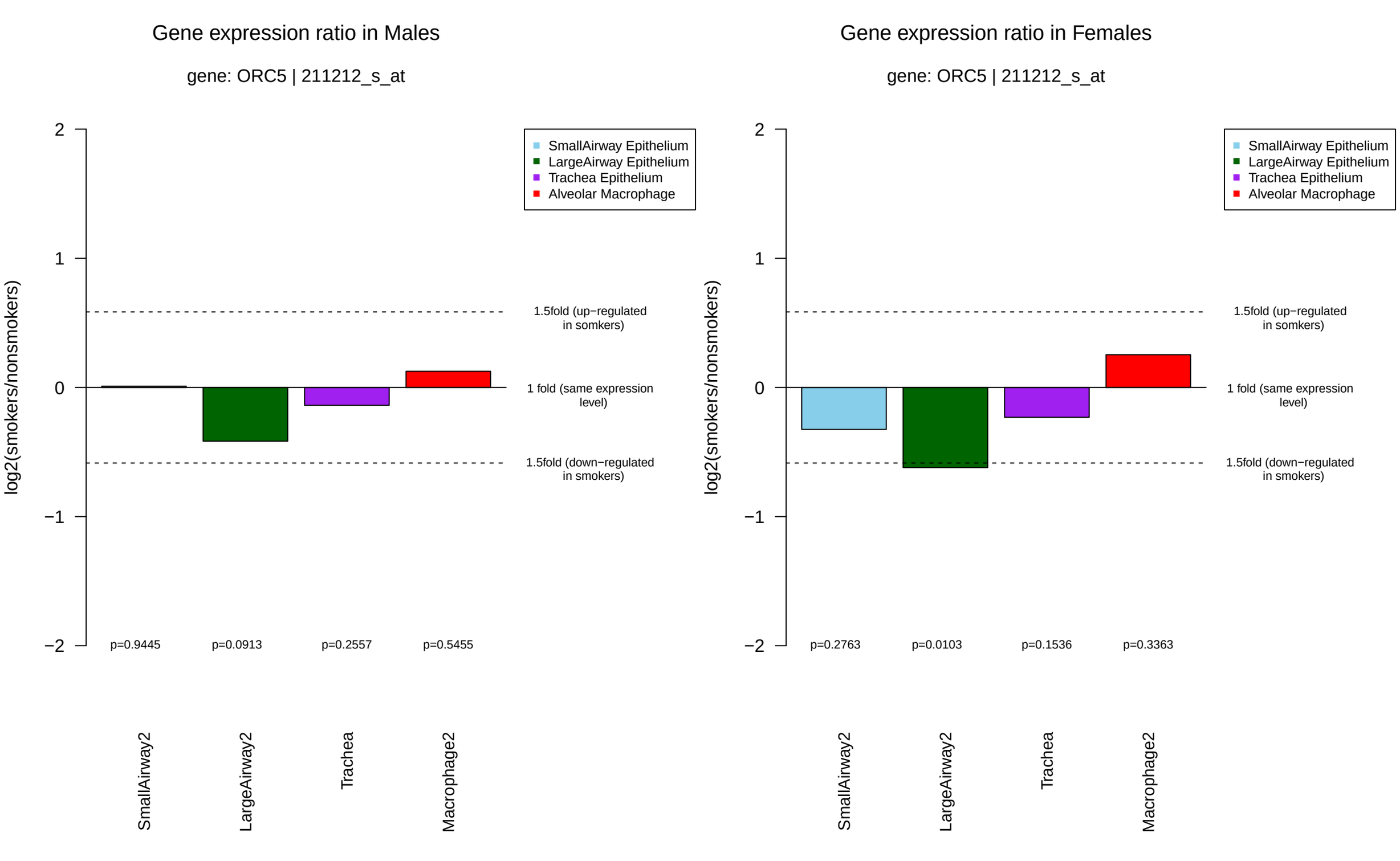
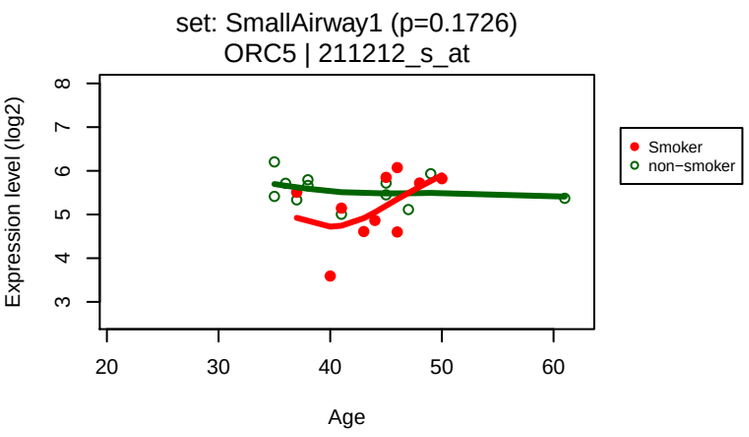
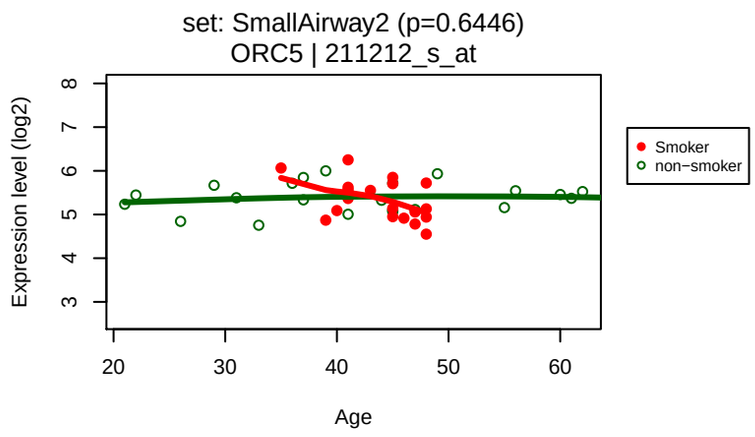
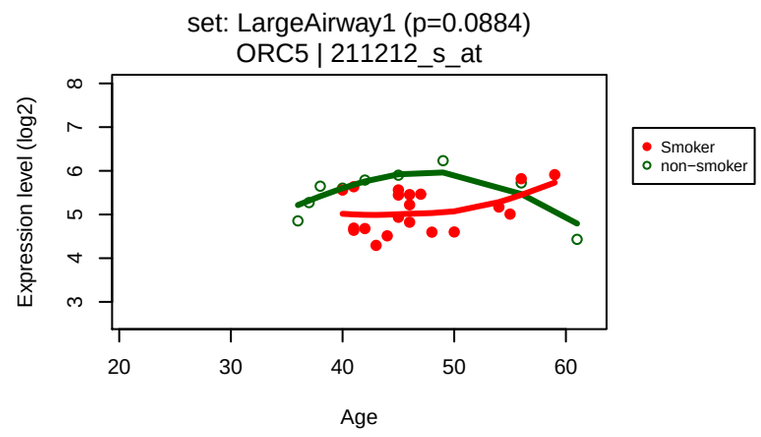

Gene expression of smokers and nonsmokers
gene: ORC5 | 211212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1726
p=0.6446
p=0.2843
p=0.0884
p=0.003
p=0.2202
p=0.6913
p=0.3222
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

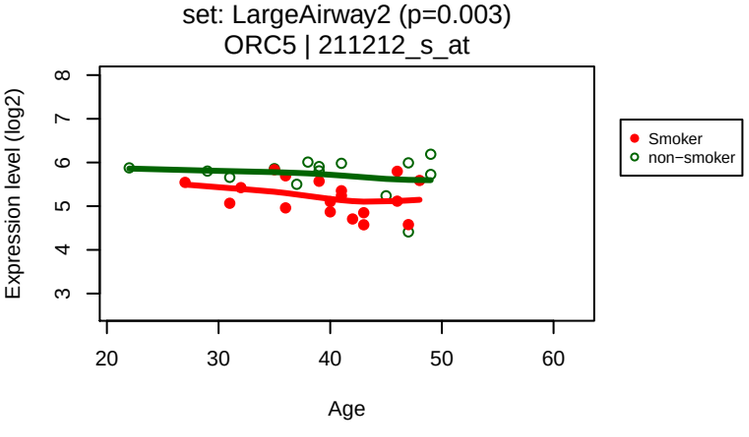

Gene expression ratio (smokers/nonsmokers)
gene: ORC5 | 211212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1726
p=0.6446
p=0.2843
p=0.0884
p=0.003
p=0.2202
p=0.6913
p=0.3222
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

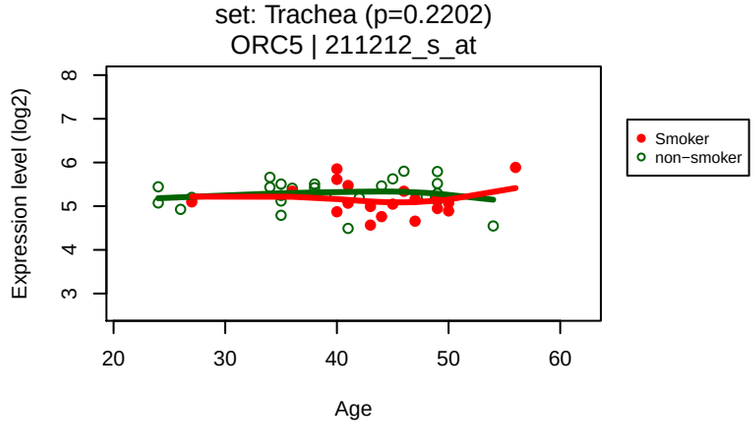

Gene expression ratio in Males
gene: ORC5 | 211212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9445
p=0.0913
p=0.2557
p=0.5455
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ORC5 | 211212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2763
p=0.0103
p=0.1536
p=0.3363
SmallAirway2
LargeAirway2
Trachea
Macrophage2

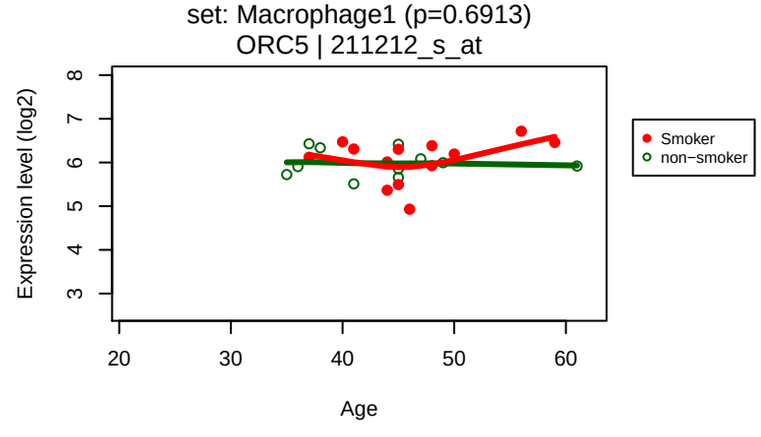

set: SmallAirway1 (p=0.1726)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker

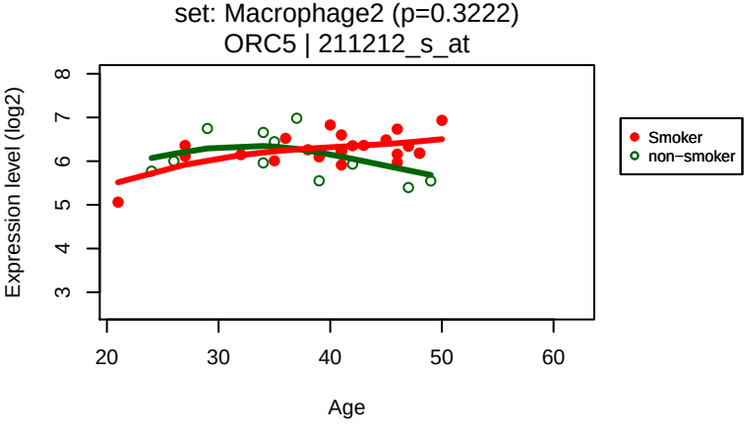

set: SmallAirway2 (p=0.6446)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker

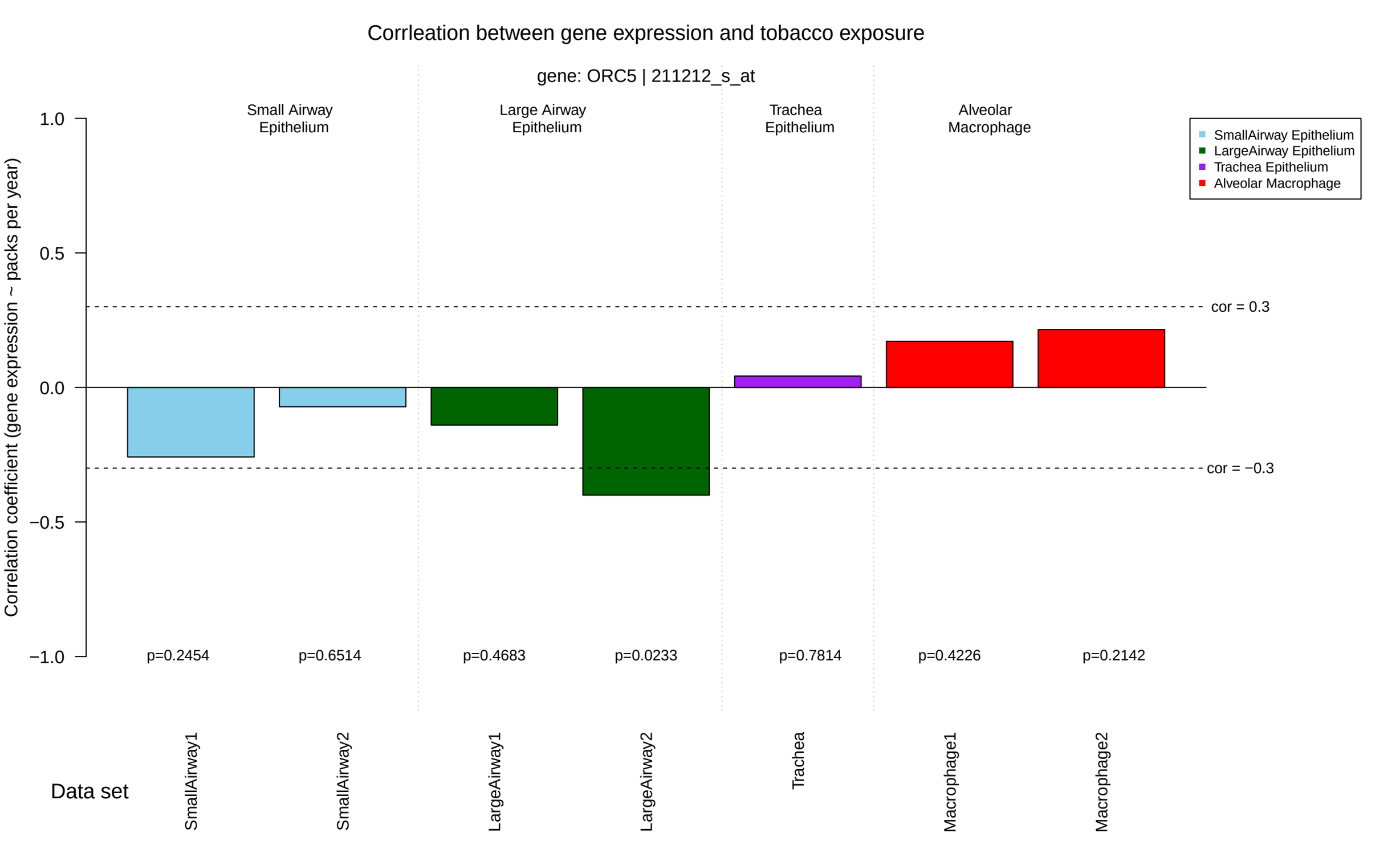

set: LargeAirway1 (p=0.0884)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.003)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2202)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6913)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3222)
ORC5 | 211212_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ORC5 | 211212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2454
p=0.6514
p=0.4683
p=0.0233
p=0.7814
p=0.4226
p=0.2142
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2